

Ch 2 Statistical learning

Distribution:

Ch 3

Terminology: Supervised learning. Y category, classification Ch 4

weight ~~MPG~~ $Y = f(X) + \epsilon$ vector

Given $X = (X_1, X_2, \dots, X_p) = p$ -indep vars

MPG $Y =$ dependent variable

find $\rightarrow f =$ function connects X & Y $\rightarrow$ Approx by line in linear regression

Data limits $\rightarrow \epsilon =$ error (mean)

CS/ML Stats Goal is to relate X & Y for:
only $\rightarrow$ ① Prediction (new X , predict Y)
 $\rightarrow$ ② Inference (X - Y connection) & Ch 3

Prediction: Approximate f by $\hat{f}$ Many options
assum $\odot$ f line. $Y \approx \hat{Y} = \hat{f}(X)$ line in linear region

New cars, predict MPG given weight.

Error analysis: Given unseen data X , how good is $\hat{Y} \approx Y$?
Computed expected value.

Review.

11 min
Stat 245

dist. $\downarrow$

Reducible (make error smaller) Irreducible (data limit) error

$$E[(Y - \hat{Y})^2] = \dots = (f(x) - \hat{f}(x))^2 + \text{Var}(\epsilon)$$

$\uparrow$ error $\uparrow$ simplify using defs + props

ϵ rand var

10 mins
Stat 245
recap

Probability basics: Example 1 roll of fair die for discrete case

Sample space S is the set of all possible outcomes

Lo Ex: $S = \{1, 2, 3, 4, 5, 6\}$

An event $E \subset S$ is any subset of S

Lo Ex: $E_2 = \{2\}$, $E_{\text{even}} = \{2, 4, 6\}$.

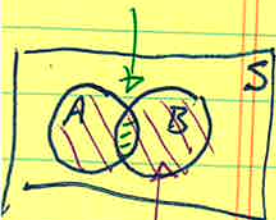
The probability of event E (equally likely outcomes)

$$P(E) = \frac{|E|}{|S|}$$

$\swarrow$ number of outcomes

$$\text{Lo } P(E_{\text{even}}) = \frac{|E_{\text{even}}|}{|S|} = \frac{3}{6} = \frac{1}{2}$$

$A \cap B$



$A \cup B$

Probability properties: $0 \leq P(E) \leq 1$

$$P(S) = 1$$

$$P(E^c) = 1 - P(E)$$

union (or)

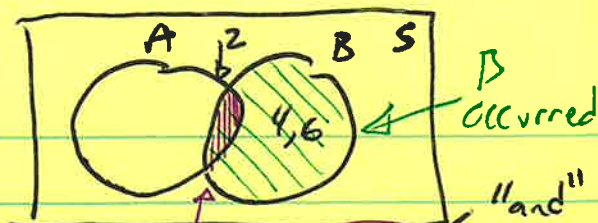
complement, not in E , Double count.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A & B independent, $P(A \cup B) = P(A) + P(B)$ 1782

The conditional probability

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{|A \cap B|/|S|}{|B|/|S|} = \frac{P(A \cap B)}{P(B)}$$



Ex ~~$P(A|B)$~~ $P(E_2 | E_{\text{even}}) = \frac{P(E_2 \cap E_{\text{even}})}{P(E_{\text{even}})}$

$$= \frac{P(E_2)}{P(E_{\text{even}})} = \frac{1/6}{1/2} = 1/3.$$

If A & B are independent,

$$P(A \cap B) = P(A) \cdot P(B) \quad \left(\begin{array}{l} \text{Rearranged} \\ P(A|B) = \frac{P(A \cap B)}{P(B)} \\ \hline P(A) \end{array} \right)$$

A discrete random variable X assigns a number to possible outcomes in S .

Ex: $S = \{ \cdot, \cdot, \cdot, \dots, \cdot \} = \{1, 2, \dots, 6\} \quad (X)$

D.R.V. X allows use of math & functions on events.

A probability mass function gives formula for outcome occurrence.

$$P(X) = P(X = x) \quad \left(\begin{array}{l} \text{leads} \\ \text{to expected.} \end{array} \right)$$

Read - Ch 1/2

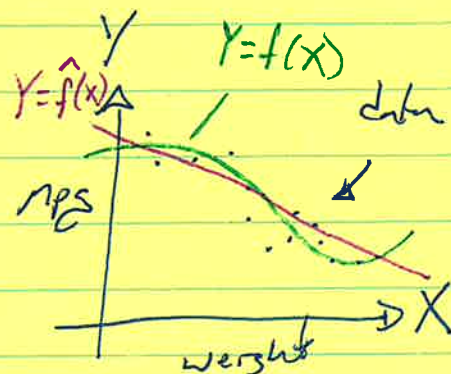
Ch 2 - Statistical learning

Recap: ^{Basis of S.L.} Probability review

S.L. supervised learning

$$Y = f(X) + \epsilon$$

$\uparrow$ dep var $\uparrow$ indep var
 $\nwarrow$ unknown fun $\nwarrow$ error



Approximate: $Y \approx \hat{Y} = \hat{f}(X)$ ^{approx fun} $\hat{f}$ ^{given data}
 ① Prediction ② Inference ($X - Y$)

How good is our approximation?

If $X, Y, f, \hat{f}$ are fixed & ϵ varies,

"Avg accuracy" $E[(Y - \hat{Y})^2] = ?$

② Probability distrib: $P(\epsilon) = \frac{|\epsilon|}{|S|}$ ^{equally likely outcomes}
 (Die roll $S = \{1, 2, \dots, 6\}$
 $E_2 = \{2\}$)

Discrete random variable X ,

(Prob mass fun)

$$p(x) = P(X=x) \quad (Ex, P(2) = P(X=2) = \frac{1}{6})$$

Expected value: $E[X] = \sum_x x p(x)$ ^{$p(x) = \frac{1}{6}$} (long term avg)

$$Ex \quad E[X] = \sum_x x p(x) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6}$$

$$= \frac{1}{6}(1+2+\dots+6) = \frac{21}{6} = \frac{7}{2} = 3.5$$

Avg X

Long term avg

Pg 1
9/4

Generalize: $E[g(X)] = \sum_k g(x) \cdot P(X)$ (avg of $g(x)$)

Variance: $\text{Var}(X) = E[(X - \mu)^2]$, $\mu = E[X]$

Ex $\text{Var}(X) = E[(X - 3.5)^2] = \dots = 2.92$ (avg variance of X from middle, avg square distance from middle)

Standard deviation: $\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$
Same units as X

Ex $\text{SD}(X) = \sqrt{2.92} = 1.7$

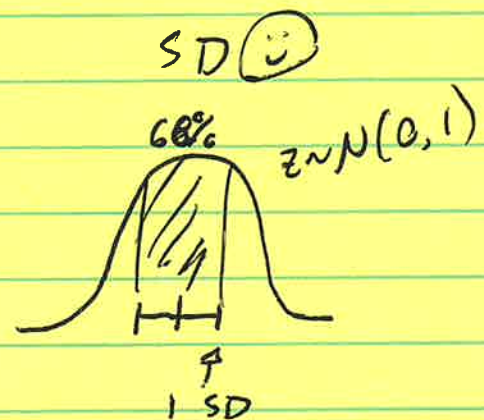
Var vs SD
Var 😊

Mathness $\mu = E[X]$

$\text{Var}(X) = E[(X - \mu)^2]$

= ...

$= E[X^2] - (E[X])^2$



1-2-3 rule, 68%-95%-99%

$$E[X] = \sum_x \underline{\underline{X}} P(X) \quad (E[X_1 + X_2] = E[X_1] + E[X_2])$$

$$E[cX] = c E[X]$$

Accuracy continued: $f, \hat{f}$ fixed WRT X, Y , ϵ can vary.

$$E[(Y - \hat{Y})^2] = E[(f(X) + \epsilon) - \hat{f}(X)]^2$$

$$= E[(f(X) - \hat{f}(X)) + \epsilon]^2$$

$E[\epsilon] = 0$
mean zero

$$= E[(f(X) - \hat{f}(X))^2 + 2\epsilon(f(X) - \hat{f}(X)) + \epsilon^2]$$

$$= E[\underbrace{(f(X) - \hat{f}(X))^2}_{\text{const}}] + E[2\epsilon(f(X) - \hat{f}(X))] + E[\epsilon^2]$$

$$= (f(X) - \hat{f}(X))^2 + 2(f(X) - \hat{f}(X)) E[\epsilon] + E[\epsilon^2]$$

$$= (f(X) - \hat{f}(X))^2 + \text{Var}(\epsilon)$$

2 parts

$$\text{So, } E[(Y - \hat{Y})^2] = (f(X) - \hat{f}(X))^2 + \text{Var}(\epsilon)$$

↑
expected
error

↑
~~Irreducible~~
error
(control $\hat{f}$ to)
reduce overall
error.

↑
Irreducible
error (decided
by data
 X, Y)

Result in overfitting or underfitting.

How to measure $Y \approx \hat{Y}$ accuracy?
- 2 ways, train error + test error.

Train set is used to fit $\hat{f}$ to known data. X, Y output (number or class)

$$\text{Train set} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

$$\text{vectors } x_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T \quad \begin{pmatrix} p \\ \text{indep} \\ \text{vars} \end{pmatrix}$$

Test set is used to assess $\hat{f}$ on unseen data.

$$\text{Test set} = \{(x_{01}, y_{01}), \dots, (x_{0k}, y_{0k})\}$$

2 types of $\hat{f}$ to train:

(i) Parametric

Ch 3/4 $\hat{f}(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$ (linear regression)

β_i parameters to fit.

(ii) Non parametric

Ch 4+ Rule based $\hat{f}$, no formula.

Many ways to measure accuracy. Standard is mean squared error (MSE). $\hat{f} \approx f$

Train: $MSE_{train} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$

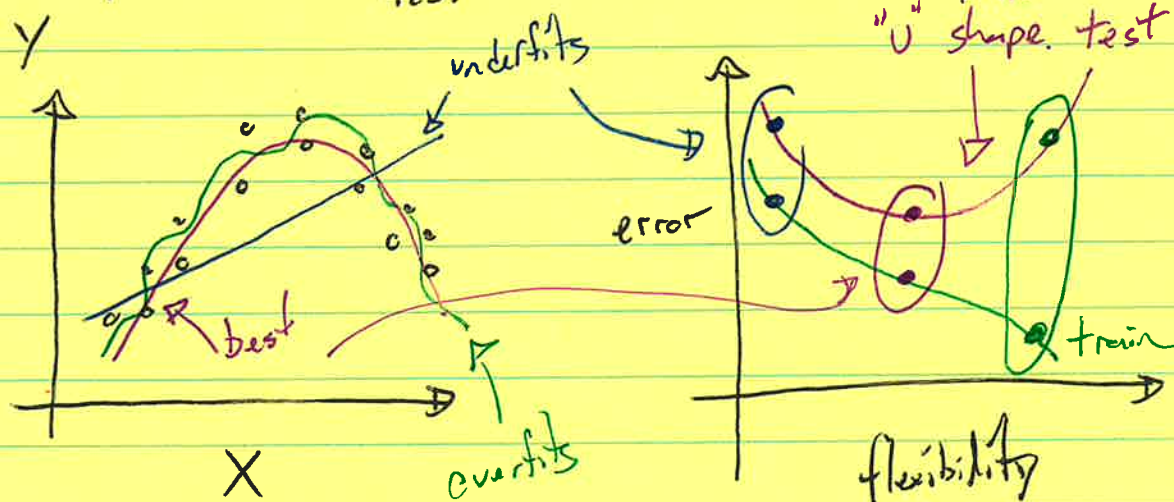
train samples n

Test: $MSE_{test} = \frac{1}{K} \sum_{i=1}^K (y_{oi} - \hat{f}(x_{oi}))^2$

Trade-off of train/test error.

- If decrease MSE_{train} ~ lot doesn't imply MSE_{test} also decreases.

Bias variance trade-off



No free lunch thm.